/

B

KAPASA MAKASA UNIVERSITY

SCHOOL OF APPLIED SCIENCE AND OPEN LEARNING

Department of Information and Communication Technology

FINAL EXAM

Course Title: Operating Systems Security

Course Code: CYS 337

Date:

Duration: 3HRS

Total Marks: 40

General Instructions to Candidates:

1. Answer a compulsory question 1 and any other 3 of your choice.
2. No unauthorized materials are allowed
3. No electronic devices allowed unless specified.
4. Start each answer on a new page.
5. Write clearly and neatly.

Please Note: *Students that are caught with foreign material, copying or behave in any manner that is viewed as examination malpractice/misconduct will be subjected to a disciplinary hearing.*

CamScanner

[QUESTION 1 Compulsory]

For Windows OS especially, it is important to implement a schedule for vulnerability assessments or scanning. Windows is popular and easy to navigate, and it's a favorite with hackers. So, there are some free software available on the market for such tasks. Outline a **detailed** step-by-step procedure on how to use Hirens Boot CD for cracking Windows Admin Login Account.

(10)

[10 Marks]

[QUESTION 2]

1) List at least two tasks of an Operating System [2 Marks]
2) Give two reasons why Microsoft developed Windows 10 [2 Marks]
3) What was the main task of MS DOS in the first version Windows [2 Marks]
4) What are the two main goals of an Operating System DUMP DART [2 Marks]
5) Tabulate any difference between Windows NT and 9x [2 Marks]

(9)

[QUESTION 3]

1) Define Context Switching [2 Marks]
2) What's the purpose of APIs in Windows? [2 Marks]
3) Explain why its difficult for an application to communicate with Hardware [2 Marks]
4) In what context does Windows Operating System use Exceptions? [2 Marks]
5) At what point in time was Windows taken into use as an Operating System [2 Marks]

(10)

[QUESTION 4]

1) What is Linux and why is popular? [2 Marks]
2) What is the difference between home directory and working directory? [2 Marks]
3) What is the difference between static library and dynamic library? [2 Marks]
4) Outline the differences between internal and external commands [2 Marks]
5) What is NFS and its purpose? [2 Marks]

(9)

g++ sokonalysis.cpp -lhss1 -o sokonalysis [#include "GMPX/gmpxx"]
gmpxx Static

g++ sokonalysis.cpp -o sokonalysis [#include <cmath>]

APPROXIMATE

Q1 = 10

Q2 = 7

Q3 = 10

Q4 = 4

33

TOTAL = 33 x (5 = 165)

CA = 33

TOTAL = 38

CA = 33

71

[QUESTION 5]

1) With a well outlined step by step procedure of your preference between Windows OS and Android OS in relation to security. Which one of the two is easier to crack ? [2 Marks]

2) What is Android OS and why is it one of the most happening Operating Systems today? [2 Marks]

3) With well tabulated reasons on vulnerability scanning between Linux OS and Android. Which one of the two Operating Systems is more secure? [2 Marks]

4) Outline two core differences between Android Operating System and Windows Operating [2 Marks]

5) List at least two other platforms that use Android Operating System apart from Mobile Phones. [2 Marks]

CamScanner